Executive Summary: ClaimShield

Kapil Rathod, Kushal Joseph Vallamkatt, Aneesh Shamraj

1 The Problem: A Broken Billing System

Studies show up to 80% of medical bills contain errors, and most patients never catch them. It's not their fault. Medical billing is complicated with CPT codes, modifiers, and clinical jargon inaccessible to the average person. The result? Patients overpay millions annually through upcoded services, phantom charges, and surprise bills they have no way to decode.

The problem runs deeper than individual overcharges. Billing fraud is systemic. Inflated codes and unbundled procedures cost insurers millions every year, and these losses ultimately get passed back to patients through higher premiums.

Is this really a problem? We compiled real medical billing errors reported by everyday Americans: see for yourself.

ClaimShield changes that, by bringing transparency, automation, and AI-driven accountability to a system long overdue for modernization.

2 How ClaimShield Works

ClaimShield is a full-stack application (Next.js frontend, FastAPI backend) powered by the OpenRouter API operating in two modes:

1. **Patient Mode**: Upload a **medical bill** and **clinical SOAP notes**. An AI agent cross-references these documents line-by-line to flag upcoding, phantom charges, and price gouging (see Fig. 1). Every flag is validated against a local vector database of 1164 official CPT codes and Medicare rates, so the AI cites sources, not guesses. A structured appeal letter can be generated instantly, with tone adjustable from *Collaborative* to *Legal Threat*.
2. **Insurer Mode**: An observability dashboard aggregates claim data by zip code, surfacing fraud hotspots and flagging providers whose billing patterns deviate significantly from Medicare standard costs.

3 AI That Doesn't Guess

Most AI hallucinates. In healthcare billing, a hallucinated code isn't just wrong: it's a liability. ClaimShield can't afford to guess, so we built it not to. Every anomaly flagged is grounded in a verified source, every step of the way:

1. **Document Parsing & OCR**: PyMuPDF and Pytesseract digitize uploaded bills and clinical SOAP notes, extracting both standard text and bounding-box spatial data to preserve document structure.
2. **LLM Auditing**: An LLM via the OpenRouter API cross-reference the bill against the doctor's own notes line-by-line, flagging upcoding, phantom charges, and price gouging with direct citations from the source documents.
3. **Grounded via RAG**: When an anomaly is detected, a vector search runs against a local database of 1,164 official CPT codes and Medicare rates. The AI is deterministically forced to use verified billing codes - it cannot invent one.
4. **Appeal Generation**: Findings are compiled into a structured, legally-informed appeal letter. The tone is fully adjustable, from *Collaborative* to *Legal Threat*, and exported as a PDF instantly, ready to send.

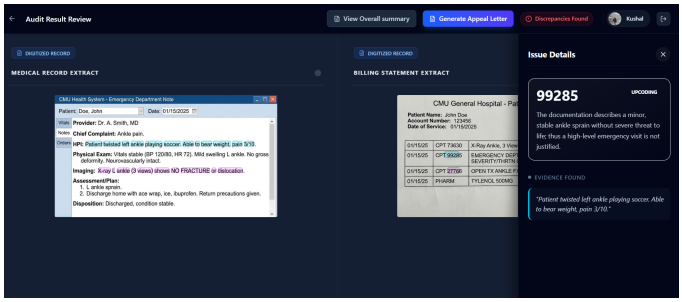

Figure 1: ClaimShield flagging an upcoded charge against clinical notes.

4 Who Benefits — and How

Patients get a fighting chance. Insurers get the visibility they've been missing. Everyone gets a system that actually works:

- **For Patients**: Medicare reference rates and plain-English findings give patients the evidence, and the confidence, to dispute inflated bills.
- **For the Overworked**: Hours of manual research and legal drafting collapsed into a single automated flow, under a minute start to finish.
- **For Insurers**: Aggregated dashboards expose organized billing fraud at scale, the kind that slips through claim-by-claim review.

Medical billing has been broken for decades. ClaimShield is the first step toward fixing it.